Earth-State Embeddings:
self-supervised representations of the observed ocean, probed against the Atlantic overturning circulation

blauewelt/earth — technical report, v5

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Abstract

We train a small self-supervised model to compress everything the observing system knows about each point of the ocean, each month, into a single vector — an *embedding* — and then ask whether those vectors, which never see any transport measurement, carry the Atlantic Meridional Overturning Circulation (AMOC). They do: a linear read-out from the embeddings along 26.5°N attains a year-blocked cross-validated correlation of $r = 0.60$ with the monthly, deseasonalized RAPID transport series — to our knowledge the first cross-validated monthly-scale figure against real RAPID data from inputs that never include RAPID. A temporal model over the frozen embeddings predicts the ocean's anomaly field beyond a year ahead (anomaly correlation 0.62 at one month, crossing the conventional 0.5 useful-skill line near four months), while the transport read-out loses skill after about one season, consistent with its wind-driven monthly variance. This revision adds a *probe ladder* — ridge, small MLP, and a supervised attention head over the *unpooled* section — and it rewrites the story of the week. Pointwise nonlinearity recovers nothing anywhere: the MLP trails the ridge on every target, for every codec. But the attention head, which sees the ∼67 section pixels individually instead of their mean, raises every codec — and the codec whose channel tokens carry 3×3 spatial patches reaches $\mathbf{r = 0.672}$ across two independently trained seeds (0.690 and 0.654), RMSE ≈2.1 Sv against a 2.79 Sv target, the best transport figure of the programme. The mechanism is thermal wind: geostrophic transport is the east-minus-west density difference across the section, and mean-pooling computes exactly the statistic in which that difference cancels. An end-to-end control then decomposes the headline: the same head over *raw* section data reaches 0.659 at matched receptive field, so pretraining's own margin on this task is +0.013 — indistinguishable from zero, and a directional +0.031 reported from the first seed alone is withdrawn here — the codec's demonstrated value lies in field prediction and transfer, not in the monthly RAPID probe. An apparent finding that expanding the input from 14 to 24–25 channels had *cost* monthly RAPID skill is retracted as a representation claim — the pooled probes were billing relocated information as lost — while the genuine gain those channels brought at the deep MOVE array at 16°N ($r = 0.206$ [0.044, 0.346], our first multi-target confidence interval excluding zero) stands.

1 Introduction

The AMOC moves on the order of 17 Sv of water northward through the Atlantic's upper kilometre ($1\,\text{Sv} = 10^6\,\text{m}^3\,\text{s}^{-1}$), carries roughly a petawatt of heat, and is directly measured at only a handful of latitudes, for only about two decades. It is the canonical example of a climate-critical quantity that is *composite*: part wind-driven (Ekman transport), part density-driven (geostrophic shear), part boundary currents. No single satellite or float sees it; it must be inferred from the state of the ocean around it.

This project asks a representation-learning question of that inference problem: *if we compress the whole observable ocean state into generic, task-blind embeddings, do climate-critical integrals like the AMOC come along for free?* The design is deliberately strict: the embedding model never sees any transport data; every read-out is a linear (ridge) regression, so any skill must already be present, linearly accessible, in the representation; and every claim is cross-validated in blocks that respect the autocorrelation of the climate system. The project also serves as a live test of a foundation-model hypothesis (§6): that a single representation trained on *all* the data — every ocean, every channel — serves specific regional tasks at least as well as one trained only on the task's own region.

2 Terminology

Terms are used throughout in the following precise senses.

Transport the physical quantity all "truth" targets measure: the volume of water a current system moves, in Sverdrups ($1\,\mathrm{Sv} = 10^6\,\mathrm{m}^3\,\mathrm{s}^{-1}$). The AMOC's strength is a transport ($\sim$17 Sv northward across 26.5°N). A *transport series* is one array's monthly record of it; none is ever a model input.

Fold one round of the year-blocked cross-validation: a calendar year serves as the test set while the probe is re-fitted from scratch on all other years. "Per fold" means exactly that re-fitting — no test month ever influences the weights that predict it.

Transport claims / field-prediction claims the project's two scoreboards, kept apart because their baselines differ. Transport claims: predicting the mooring series (k-fold r, RMSE in Sv, event capture) against the wind-only ridge. Field-prediction claims: predicting the ocean's own input fields ahead in time (chan%, rollout MSSS/ACC) against persistence and climatology.

Channel one physical input field: e.g. surface current speed, mixed-layer depth, temperature at 700 dbar, zonal wind stress. The data tensor has 12–15 channels in the first generation, 24–25 in the second, and 39 in the third (Table 1, §3.2).

Tensor generation one build of the input tensor whose channel set, grid, or time axis differs from the others enough that results are not comparable across it (§3.2). Three exist, and we name them by what defines them: the **coarse tensor** (1°, 4 Argo pressure levels; "family 1" in earlier notes), the **deep tensor** (1°, 8 Argo levels; "family 2"), and the **quarter-degree tensor** (0.25° North Atlantic, 16 Argo levels, sea-surface height added, time axis extended to 1982; "family 3"). Run names keep the short prefixes (`f3_` etc.); prose uses the descriptive names.

Pixel one ocean grid cell — $1° \times 1°$ in the first two tensor generations, $0.25° \times 0.25°$ in the third. A *pixel-month* is one pixel at one month — the atomic sample.

Channel space the space of the physical fields themselves. An error "in channel space" is an error in predicted physical values (normalized units), directly comparable with a persistence forecast of the same fields.

Embedding / embedding space the 64-dimensional vector z the encoder produces for a pixel-month, and the space of such vectors. An error "in embedding space" compares predicted to actual z; it is an internal diagnostic, since z has no physical units.

Codec the stage-1 encoder–decoder (§4); it encodes a pixel-month's channels to z and can decode z back to channels. "Codec" emphasizes that it is a learned compression.

Anomaly space all dynamic channels are expressed as departures from that pixel's own monthly climatology (computed over training years only), then normalized. This removes everything predictable from location and calendar alone.

Climatology the long-term mean for a given pixel and calendar month. As a forecast, "climatology" predicts zero anomaly — the no-skill floor every real forecast must beat.

Persistence the forecast that the anomaly stays exactly as it is now. Hard to beat at short range; decays into a poor forecast at long range. *Damped persistence* multiplies the anomaly by its own lag-h autocorrelation — the fair statistical baseline.

Patch codec a codec variant whose channel token carries the 3×3 neighbourhood of the pixel (nine values and nine observed-flags) rather than a single scalar, giving the encoder gradient vision and turning the representation from an expander into a genuine compressor (§4).

Stage 2 / temporal model a causal transformer over a pixel's sequence of frozen embeddings that predicts next month's embedding. "Frozen" means the codec's weights are not updated by stage-2 training, so improvements are attributable.

Probe a read-out fitted from frozen embeddings to a target series; the codec and stage 2 are never updated by it. Historically a ridge regression, deliberately weak; now the bottom rung of the probe ladder below. A probe measures what a representation *contains*.

channel(s)	source	record	role
surface current speed, MLD	GLORYS reanalysis	1993–	dynamics
sea-surface height ($\frac{1}{4}°$ only)	GLORYS reanalysis	1993–	geostrophy
T at 4 / 8 / 16 pressure levels	Roemmich–Gilson Argo	2004–	density structure
S at 4 / 8 / 16 pressure levels	Roemmich–Gilson Argo	2004–	density structure
wind stress τ_x, τ_y	NCEP R1	1948–	Ekman forcing
within-month τ std. dev.	NCEP R1 daily	1948–	storminess
monthly SST (optional, 1° only)	OISST v2.1	1981–	thermal memory
SST, precip climatologies (1° only)	OISST, GPCP	fixed	(measured inert; §8)

Table 1: Channels of the data tensor across its three generations (§3.2). Coarse and deep tensors: monthly, 1°, 1993–present; NA pilot window (100°W–20°E, 0–70°N; 5,787 ocean pixels) or global (44,964 pixels). Quarter-degree tensor: monthly, 0.25°, NA window, 84,405 ocean pixels, axis extended to 1982-01 (channels absent before their records enter as missing tokens), C=39. Missing data (Argo before 2004, polar gaps, pre-1993 months) is represented explicitly, never imputed.

Probe ladder three read-outs of the same frozen embeddings, increasing only in what they may see: *ridge* (linear, on the section-mean embedding), *MLP* (adds pointwise nonlinearity, same pooled input), *attention head* (adds spatial structure: one learned query attends over the unpooled per-pixel embeddings of the section). Each rung localizes where a capability comes from.

Section the row of grid cells nearest a mooring array's latitude, clipped to the array's longitude span (e.g. RAPID: 26.5°N, 80°W–13°W). Probes read the mean embedding along the target's section.

Holdout years entire calendar years (2009, 2017, 2023) excluded from all training; with a held-out mid-Atlantic longitude band they form the blocked validation splits. Random splits are never used — adjacent months are too correlated to count as independent.

k-fold (year-blocked) every calendar year takes one turn as the test set while the probe is fit on the others; the assembled out-of-fold predictions are scored once. Confidence intervals come from a bootstrap that resamples whole years (a *block bootstrap*).

chan% held-out next-month prediction error reduction versus persistence, in channel space, in anomaly space: $100\,(1 - \mathrm{MSE}_{\mathrm{model}}/\mathrm{MSE}_{\mathrm{pers}})$. Comparable only within one channel set, since the persistence baseline moves when channels change.

MSSS mean squared skill score, $1 - \mathrm{MSE}/\mathrm{MSE}_{\mathrm{ref}}$ [6]; our rollout reports it against climatology, persistence, and damped persistence.

ACC anomaly correlation coefficient: the Pearson correlation between predicted and observed anomalies at a given forecast horizon. ACC = 0.5 is the conventional "useful skill" line.

RAPID / FC / MOVE / OSNAP / SAMBA the five transport truth series: the RAPID array at 26.5°N (2004–, monthly), the Florida Current cable at 27°N (1982–, daily), MOVE at 16°N (2000–2022, the *deep* western-boundary flow), OSNAP in the subpolar gyre (2014–2022), SAMBA at 34.5°S (2009–2017). None is ever an input.

3 Data

The tensor (Table 1) is built entirely from credential-free public archives. Two properties matter most. First, *missingness is information*: a pixel-month's unobserved channels enter the encoder as learned "missing" tokens rather than imputed values, so the representation knows what the observing system knew. Second, everything is in anomaly space (§2) — the founding negative result of the project was that a model trained on raw fields wins reconstruction by memorizing the seasonal cycle while learning nothing about change.

The five truth series (RAPID, FC, MOVE, OSNAP, SAMBA) ride along as never-input targets: 240 RAPID months plus 792 additional truth months across four other latitudes.

3.1 The sources, one by one

Every input is a credential-free public archive; this subsection says what each one is, what channels it contributes, and why it is in the tensor. (The companion visualization at `blauewelt.github.io/earth` renders most of these sources as interactive globe layers with per-dataset documentation; the descriptions here follow its catalog.)

GLORYS ocean reanalysis (CMEMS). An eddy-permitting model of the global ocean constrained by satellite altimetry, SST, sea ice, and in situ profiles — the closest thing oceanography has to a best-estimate movie of the ocean since 1993. Contributes monthly *surface current speed* ($\sqrt{u^2 + v^2}$; the western boundary currents that carry the AMOC's upper limb are current-speed structures) and *mixed-layer depth* (log-transformed; deep winter mixing in the subpolar gyre is where surface waters acquire the density to sink — the "engine room" of the overturning). At 1° these come from the quarter-degree ensemble member binned down; the quarter-degree tensor reads the native grid and adds *sea-surface height*, whose meridional gradient is the geostrophic surface flow and whose boundary difference is the classical altimetry AMOC proxy [1].

Roemmich–Gilson Argo climatology. The gridded monthly temperature and salinity fields (1°, 0–2000 dbar) built from the Argo float array [21] — the backbone subsurface observing system: ∼4,000 autonomous floats, each profiling every ∼10 days. Contributes T and S at 4, 8, or 16 pressure levels (10–1900 dbar) depending on tensor generation. These are the density channels: thermal wind makes the transport an east-minus-west density difference, and the deep levels see the cold limb the surface channels cannot. Intrinsically 1°-smooth (in the quarter-degree tensor they are bilinearly upsampled and carry no sub-degree information — stated, not hidden). The record begins in 2004; earlier months are missing tokens.

NCEP/NCAR Reanalysis 1 surface wind stress. The longest-running atmospheric reanalysis [22], daily since 1948, on a Gaussian grid regridded to the tensor. Contributes monthly-mean τ_x, τ_y (Ekman transport, the wind-driven part of the AMOC's monthly variability — and hence also the source of the wind-only baseline every codec must beat) and the *within-month standard deviation* of the daily stress (storminess: buoyancy loss in storms drives convection in a way monthly means erase). Its length is what lets the quarter-degree tensor's axis start in 1982.

NOAA OISST v2.1. Daily 0.25° SST from AVHRR plus in situ data since late 1981 [23], monthly-meaned; optional channel 15 of the deep tensor, and (as a fixed 1991–2020 climatology) one of the two deliberately inert channels of §8.

GPCP v2.3 precipitation climatology. The satellite–gauge merged precipitation record [24]; enters only as the second inert climatology channel of §8.

Truth arrays (never inputs). Five basin-scale transport series, each a moored or cable observing system: *RAPID–MOCHA* 26.5°N [25], the flagship full-depth AMOC record (2004–, our 240 labelled months); the *Florida Current submarine-cable transport* [26], the lowest-latency AMOC component, daily since 1982 — the decade 1982–92 is usable truth only for the quarter-degree tensor's extended axis; *MOVE* 16°N [27], the longest deep-limb record (southward NADW flow, 2000–); *OSNAP* [28], the subpolar array where overturning variability originates (2014–); and *SAMBA* 34.5°S [29], the only sustained South Atlantic array (2009–, anomalies only).

3.2 Tensor generations, and why they may not be compared

The tensor is rebuilt from the fetcher scripts at the start of every training job, so a change to a fetcher silently changes the input of every subsequent run. Sampling 8 Argo pressure levels instead of 4 took the tensor from 14 channels to 24 (25 with monthly SST) without any dispatch asking for it. Every result below therefore belongs to one of three *tensor generations* — the **coarse tensor** (1°, 4 Argo levels), the **deep tensor** (1°, 8 levels), and the **quarter-degree tensor** (0.25° NA, 16 levels, SSH, axis from 1982) — and cross-generation comparison is not licensed: the persistence baseline that chan% is measured against moves with the channel set, the wind-only ridge baseline moves with the grid (0.531 on the 1° tensors, 0.568 on the quarter-degree), and, as §7.5 shows, so does the meaning of the transport probe. (Earlier project notes call these "channel families" 1–3; run names keep the shorter prefixes.) Channel counts in this report are read from each checkpoint, never from a run's name — a run directory carried a misleading name for an hour before this was enforced, and every run now ships a provenance manifest recording the tensor it actually saw.

4 Methods

Stage 1 — the codec. A per-pixel masked autoencoder producing $z \in \mathbb{R}^{64}$ per pixel-month, trained on all pixel-months outside the holdout blocks. The pilot is 0.92M parameters; a 10.25M variant (320×8, the size the Chinchilla-style arithmetic of the Compute paragraph anchors for the current tensor) is training at time of writing. Checkpoints carry their architecture, and every loader reconstructs from it.

What exactly enters the encoder. The encoder is a small transformer (128×4 in the pilot, 320×8 in the 10M variant) over $C{+}2$ tokens for one pixel-month:

- **One token per channel**. If the channel is observed and not masked, the token is a linear projection of its scalar value (in anomaly space: the departure from that pixel's own monthly climatology, standardized) plus a learned per-channel identity embedding. If the channel is *masked for training*, the value is replaced by a learned `mask` token; if it is *genuinely unobserved* — Argo before 2004, polar gaps — by a distinct learned `missing` token. "I am hiding this from you" and "the observing system never measured this" are different facts, and the distinction measurably matters.
- **One context token**: a linear projection of $[\sin(2\pi m/12),\ \cos(2\pi m/12),\ \text{lat}/90,\ \text{lon}/180]$ — season and position. Never masked. This token is why static climatology input channels carry zero information (§8).
- **One CLS token**, whose final hidden state is projected to $z \in \mathbb{R}^{64}$.

The decoder never sees the inputs — it answers *queries* (channel, Δlon, Δlat, Δmonth) from z alone, with offsets up to ± 3 cells and months, so the training signal is reconstruction of masked channels *plus* prediction of spatial and temporal neighbours.

The patch codec. At 14 channels a pixel-month carries on average 10.4 observed values, so a 64-dimensional z is an *expander* ($\sim 6\times$), as in masked language models; the information bottleneck is the masking objective, not the dimensionality. The patch variant gives each channel token its 3×3 neighbourhood — nine values and nine observed-flags, projected by $\text{Linear}(2p^2, d_{\text{model}})$ — so that at 24 channels roughly 160 observed values enter one 64-dimensional vector, a genuine 2.5:1 compression. Longitude wraps (the globe is periodic in x) and latitude clamps with out-of-range rows marked unobserved; the centre cell defines whether a channel counts as observed, and reconstruction targets remain centre values, so the loss is unchanged and $p{=}1$ checkpoints load identically. The motivation is physical as well as informational: thermal wind *is* a density gradient, and a gradient does not exist within a single pixel.

Stage 2 — the temporal model. A causal transformer (default $d_{\text{model}}{=}96 \times 3$ layers, 0.35M parameters; capacity sweep in Fig. 6) over each pixel's 24-month embedding sequence, predicting z_{t+1}; scored after decoding to channel space. The codec stays frozen.

Probes — the ladder. The base instrument is a ridge regression from section-mean embeddings to each truth series, deseasonalized, regularization chosen on a training tail — never on test data; verification is year-blocked k-fold with block-bootstrap confidence intervals. Two stronger read-outs sit above it, behind the same folds and the same inner-tail discipline. A small MLP ($64 \to 32 \to 1$, weight decay, early stopping, three seeds per fold) tests whether skill is present but not *linearly* accessible. An attention head tests whether skill is present but not accessible from the *pooled* section: every (pixel, month) embedding enters as a token with a longitude encoding, one learned query cross-attends over them, and a two-layer read-out maps the pooled vector to transport ($\sim$25k parameters, weight decay 10^{-2}, token dropout, three seeds per fold). The codec stays frozen throughout — these are probes, not fine-tuning. The design principle: each rung adds one capability, so a gap between rungs is attributable to that capability alone. Beneath the ladder sit two end-to-end *controls*: the identical head fed raw section values and observed-flags instead of embeddings, at single-pixel and at 3×3 receptive fields — pretraining is then the only manipulated variable (§7.3).

Rollout. For horizons $h = 1 \ldots 12$: initialize with true embeddings, predict one month, feed the prediction back, repeat — scored in channel space against climatology, persistence, and damped persistence, and probed for the AMOC at each horizon.

Compute. Runs through 2026-08-07 were trained on GitHub-hosted 4-core CPU runners at ∼1.3 steps/s. Since then three rented consumer GPUs (RTX 4090, ∼\$0.24/h each) are attached as self-hosted runners at ∼124 steps/s each — a factor of 95 per box, which turns a 30k-step schedule from six hours into four minutes and made the controlled comparisons of §7 affordable. The raw source data (∼13 GB) is mirrored as a GitHub release so a fresh box seeds its cache from a CDN rather than the origin archives, one of which (the Argo host) times out under load. Nothing about the model or protocol changed with the hardware. Chinchilla-style anchor for sizing: the current tensor carries ∼270M observed values per epoch; at the ∼20:1 data-to-params ratio that suggests a ∼13M-parameter codec — 14× the pilot, and the reason the 10M variant exists. The arithmetic is redone whenever the data changes.

5 Metrics and their statistical power

Full definitions are in the glossary; the reliability ordering is the substance. **chan%** is decision-grade within a channel set: seed spread ±0.5 points, so differences ≥ 2 points are real. The **fixed-holdout probe** r (36 held-out months, effective sample ≈ 15 after autocorrelation) has a 95% CI of ±0.57 — a compass, never a verdict; the same configuration has drawn 0.09 and 0.33 on two seeds. The **year-blocked k-fold** tests all 240 RAPID months once each (SE ≈ 0.10) and is the only probe number we argue from. The most instructive failure: a d_z=128 codec set the all-time fixed-holdout record (0.528) while the k-fold read 0.166 — the coarse instrument flatters widest exactly where overfitting is easiest.

6 Related work

Three literatures border this project; we take them in turn. (The repository file `ml/LITERATURE.md` carries the full survey and is updated more often than this section.)

AMOC reconstruction from observables. The classical line regresses the RAPID transport on physically chosen observables. Frajka-Williams [1] reconstructs the MOC from satellite sea-level anomaly plus the cable and Ekman terms and explains >90% of variance ($r \approx 0.95$) — but after 18-month low-pass filtering, with in-sample calibration; Sanchez-Franks et al. [2] reach $r = 0.83$ from altimetry via a thermal-wind argument under the same filtering; Worthington et al. [8] fit boundary densities to the upper-mid-ocean transport ($R^2 = 0.78$ monthly, held-out $r \approx 0.75$ on a single 2017–18 test year). Fingerprint methods [9] recover multi-decadal AMOC trends from SST patterns but no monthly signal. The machine-learning line trains almost exclusively on simulations: Solodoch et al. [3] report $r \approx 0.98$ for a CNN reading OBP/SSH/SST/wind strips — inside the laminar 1° ECCO state estimate; in eddy-resolving models the same task drops to explained variance 0.44–0.74 [4, 5], which we read as the honest difficulty scale. Application to real RAPID data at monthly cadence with blocked cross-validation is, as far as we can determine, absent from the literature. Our pooled-ridge $r = 0.60$ — monthly, deseasonalized, real data, year-blocked k-fold, no RAPID-derived inputs — sits where the eddying simulations predict it should (the attention-head read-out reaches 0.65–0.69 across two seeds, with the attribution caveats of §7.3), and no published surface-to-OSNAP reconstruction exists for our OSNAP null to contradict.

6.1 Why these numbers are not comparable

Table 2 spans $r = 0.44$ to $r = 0.98$, and almost none of that spread is model quality. Four mechanisms, in descending order of how much they matter.

study	inputs	target & filtering	protocol	skill
Frajka-Williams [1]	altimetry + **cable** + Ekman	RAPID MOC, 18-mo low-pass	in-sample	$r \approx 0.95$
Sanchez-Franks [2]	altimetry (thermal wind)	RAPID, deseas. + 18-mo	in-sample	$r = 0.83$
Worthington [8]	boundary densities	RAPID *UMO*, monthly	1 test year	$R^2 = 0.78$; $r \approx 0.75$
Caesar [9]	subpolar SST fingerprint	AMOC index, 20-yr LOWESS	model-calibrated	trend only
Solodoch [3]	OBP+SSH+SST+τ	**ECCO** MOC, monthly	chrono. 70/30, *simulation*	$r \approx 0.98$
Meng [4]	SSH+OBP maps	eddying-model MOC	*simulation*	skill ≈ 0.44
Wölker [5]	virtual Argo + τ + cable	VIKING20X 26.5°N	*simulation*	$R^2 \approx 0.74$
this work (ridge)	14-ch embedding, **no transport**	**real** RAPID, monthly deseas.	year-blocked k-fold	$r = 0.602$ [0.46, 0.73]
this work (head)	24-ch embedding, **no transport**	**real** RAPID, monthly deseas.	year-blocked k-fold, 2 seeds	$r = 0.672$
same, under their filter	—	+18-mo low-pass	*as above*	$r = 0.33$–0.75
classical inputs re-scored in OUR protocol (§6.2) — same 227 months, same folds				
wind only	section τ_x, τ_y	real RAPID, monthly	year-blocked k-fold	$r = 0.401$ [0.27, 0.56]
cable only	**Florida Current transport**	real RAPID, monthly	year-blocked k-fold	$r = 0.254$ [0.15, 0.36]
cable + wind	**two of RAPID's three terms**	real RAPID, monthly	year-blocked k-fold	$r = 0.566$ [0.45, 0.68]

Table 2: **Joint results table.** Every row is a defensible number in its own frame; the frames differ so much that the rightmost column should not be read as a ranking. §6.1 explains each difference and quantifies the largest. Bold marks the two design choices that most change the task: whether a *transport measurement* is among the inputs, and whether the target is the *real* ocean or a simulation.

1. Some inputs already contain the answer. RAPID does not measure one thing; it sums three: the Florida Current transport, the wind-driven Ekman transport, and the upper-mid-ocean geostrophic transport. Frajka-Williams [1] takes the *cable* record (which *is* the Florida Current term, measured) and Ekman (from wind) as inputs, and reconstructs only the third. Two of the three addends are handed to the model; the reported correlation is substantially the correlation of a partial sum with its own total. Wölker [5] likewise includes cable transport. Our inputs contain no transport measurement of any kind — not the cable, not Ekman as a transport — so we are inferring all three terms from the ocean state. These are different problems, and the harder one should be expected to score lower.

2. The 18-month filter removes 83% of the variance and 87% of the degrees of freedom. This is the difference most often glossed as a methodological detail, so we measured it on the RAPID series itself. Monthly deseasonalized RAPID has $\sigma = 2.79\,\mathrm{Sv}$, a lag-1 autocorrelation of 0.32, and an integral timescale of 3.5 months — so its 240 months carry ≈68 effective degrees of freedom. After an 18-month low-pass, the integral timescale becomes 25 months and the effective sample size falls to ≈**9**, while the retained variance is **17%** of the original. Two consequences. First, the filtered task is easier: what survives is the slow, heat-content-like component that altimetry and density observables track well, while the wind-driven month-to-month variance — the part we are scored on — has been removed. Second, the filtered statistic is barely constrained: $r = 0.95$ on 9 effective points carries a 95% confidence interval of roughly $[0.77, 0.99]$, and $r = 0.83$ one of $[0.37, 0.96]$ — intervals wide enough to overlap our unfiltered number. (As a check on the arithmetic: the same calculation gives $[0.42, 0.73]$ for our $r = 0.60$ at 68 effective points, against the $[0.46, 0.73]$ our block bootstrap reports independently.)

We can also show the instability empirically rather than asserting it. Two codecs of *identical* configuration differing only in training seed score 0.543 and 0.531 monthly — a gap of 0.012 — but 0.449 and 0.327 once 18-month filtered, a gap of 0.122, ten times larger. Across our runs the filtered correlation ranges from 0.33 to 0.75 while the monthly one stays within 0.53–0.62. We report the filtered number for every run (it is in the repository's leaderboard), but we decline to headline a statistic whose seed noise is an order of magnitude larger than the effect anyone wants to measure.

3. In-sample calibration is a fit, not a prediction. The classical reconstructions [1, 2] tune their coefficients on the same period over which skill is then reported; Caesar [9] calibrates against models. No held-out set exists, so the number answers "how well can this functional form describe these years" rather than "what would it say about a year it has not seen". Worthington [8] is the honourable exception in the classical line, holding out 2017–18 — one test year, where our protocol re-fits from scratch for every year in turn. Our year-blocked k-fold is the strictest arrangement here and costs us skill relative to every in-sample row, by construction.

4. A simulated ocean is a different ocean. Solodoch's $r \approx 0.98$ [3] is obtained inside ECCO, a 1° state estimate whose dynamics are laminar and whose surface fields and MOC are generated by the same model — the mapping to be learned is close to deterministic and free of observational noise. The moment the same task is posed in eddy-resolving simulations it falls to 0.44–0.74 [4, 5], and the real ocean adds sparse, irregular sampling on top of eddies. We read the eddying-model range, not the laminar one, as the relevant difficulty scale — and our 0.60–0.67 on real observations sits inside it.

6.2 Making them comparable: the classical inputs under our protocol

A like-for-like comparison requires real RAPID data, monthly resolution without low-pass filtering, no transport measurement among the inputs, and an out-of-sample protocol; no published result meets all four. But the comparison can be rescued from the other end. Rather than dress our number up in the literature's conventions, we put the literature's *inputs* through our protocol — same target, same deseasonalization, same year-blocked folds, same block bootstrap — so that the rows differ only in what the model is allowed to see. The Florida Current cable, normally one of our never-input truth series, is used as an input for these rows and these rows only (`ml/classical_baseline.py`).

The bottom block of Table 2 reports it, on the 227 months where RAPID and the cable overlap. Wind stress alone gives $r = 0.401$; the cable alone $r = 0.254$; **the two together — i.e. two of RAPID's three constituent terms, handed to the model as direct measurements — give** $r = 0.566$ [0.45, 0.68]. Our embedding, which is shown no transport measurement of any kind, reaches 0.578–0.620 under the same pooled ridge and 0.672 with the unpooled head.

We read that carefully. It does *not* say we beat Frajka-Williams: that reconstruction also uses altimetry for the third term, which this ladder omits, and it targets a filtered series. What it does say is sharper and is the number we could not get any other way: at monthly resolution, out of sample, being handed the Florida Current and the Ekman transport outright is worth about $r = 0.57$ — and a task-blind embedding of the ocean state matches it without being handed anything. The gap between that 0.57 and a published 0.95 is therefore not model quality; it is the 18-month filter and the in-sample fit, quantified. A supporting observation in the same direction: when we apply the 18-month filter to these classical rows' *out-of-fold* predictions their correlations collapse (−0.07, −0.23, +0.24), which is what a filtered statistic with nine effective points does once it is no longer allowed to see its own test years.

Ocean and climate forecasting. Our field-prediction metrics are the standard decadal-verification quantities under other names: skill vs. climatology is MSSS with a climatological reference, and we adopt ACC per lead and damped persistence from the same framework [6]. Reference points at our monthly cadence: Ham et al.'s ENSO CNN [10] holds Niño3.4 correlation > 0.5 to $\sim$17 months (the ocean's most predictable index); Taylor & Feng [11] forecast global monthly SST anomaly fields with a Unet–LSTM (RMSE 0.48°C at 1 month rising to 0.63 at 18) — the closest published analogue to our stage-2 rollouts. The daily 1/4°–1/12° neural ocean forecasters (XiHe, WenHai, GLONET) and the global weather models they descend from [12, 13] operate in a different regime (days, full fields, reanalysis-supervised) and are cited for orientation, not comparison.

Self-supervised and foundation models for Earth observation. Architecturally we are a masked autoencoder [14] over channel tokens rather than image patches. The flagship Earth-system foundation

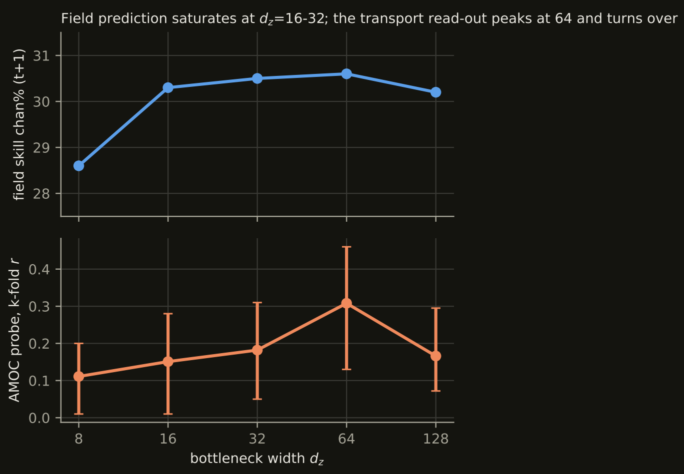

Figure 1: Bottleneck width d_z versus its two customers, on the 12-channel North-Atlantic tensor at matched training steps. Field prediction (top) saturates by d_z=16–32; the AMOC probe (bottom, with 95% block-bootstrap CIs) rises to a peak at d_z=64 and turns over at 128 — 128 ridge features against ∼220 truth months per fold overfits. d_z=64 became the working setting, a choice made when the tensor had 12 channels.

models — ClimaX [15], Aurora [16], and the land-surface line of Prithvi [17] and SatMAE [18] — evaluate mainly by full fine-tuning; frozen-representation evaluation, our entire protocol, is the exception: Prithvi WxC's anomaly-space masked pretraining with frozen backbone and small heads is our closest architectural relative (atmosphere-only), and Lehmann et al. [7] argue explicitly that frozen-backbone extension should be a standard foundation-model quality metric. AlphaEarth Foundations [19] embeds every terrestrial 10 m pixel into 64 dimensions — the same width we use — from ten data modalities, and is the clearest statement of the "embedding field" idea this project transplants to the ocean; it is terrestrial-only, and its modality list (translated to ocean: altimetry, gravimetry, scatterometry, ocean colour, sea ice) is our data-ladder shopping list. Model sizing follows the compute-optimal-scaling heuristic of Hoffmann et al. [20] adapted to observed-value counts (§4, Compute). To our knowledge no self-supervised foundation model exists for the gridded ocean interior; frozen-embedding probes to integrated circulation indices, and missingness-as-information over the historical observing system, remain unoccupied territory.

7 Results

7.1 The bottleneck: what width buys

Fig. 1: the transport read-out wants more width than field reconstruction rewards, but not unboundedly — the turnover at d_z=128 is a truth-data limit, not a representation limit. That this sweep was run at C=12 becomes important in §7.5.

7.2 The probe ranking, and what the wind already knows

Fig. 2 is the project's headline and its conscience in one picture. The wind channels moved the probe more than every architecture decision combined (0.31 → 0.60); the global codec matches the regional one exactly (0.602 vs 0.604 — the generic-embedding hypothesis holds at fixed capacity); and the wind-only baseline at 0.53 says most monthly skill is Ekman physics. The coarse tensor's embedding increment over pure wind is +0.07; its distinctive contributions are the low-frequency component (18-month low-passed r = 0.64) and event anatomy (Fig. 7). In Sverdrups: RMSE 2.23 Sv against a target standard deviation of 2.79 Sv.

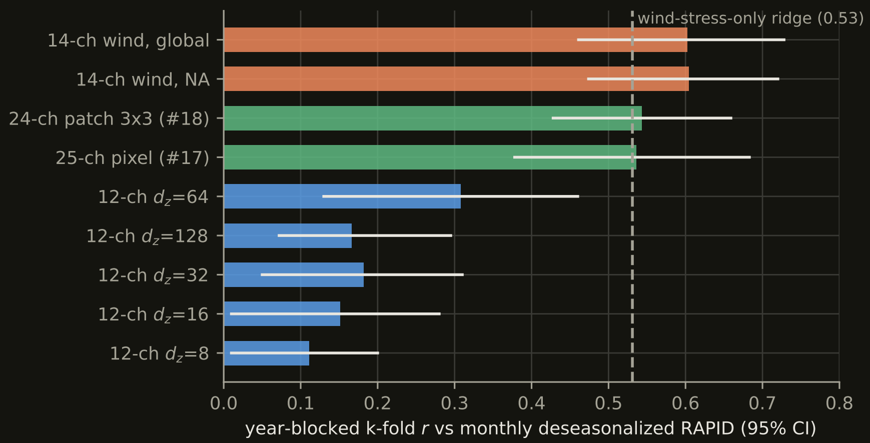

Figure 2: Year-blocked k-fold correlation with monthly deseasonalized RAPID transport, all codec generations, 95% CIs. Adding wind stress (orange) nearly doubled the defensible probe; training globally cost the Atlantic read-out nothing. The deep-tensor codecs (green) fall back onto the dashed wind-only baseline — a ridge on section-mean wind stress with no embedding at all.

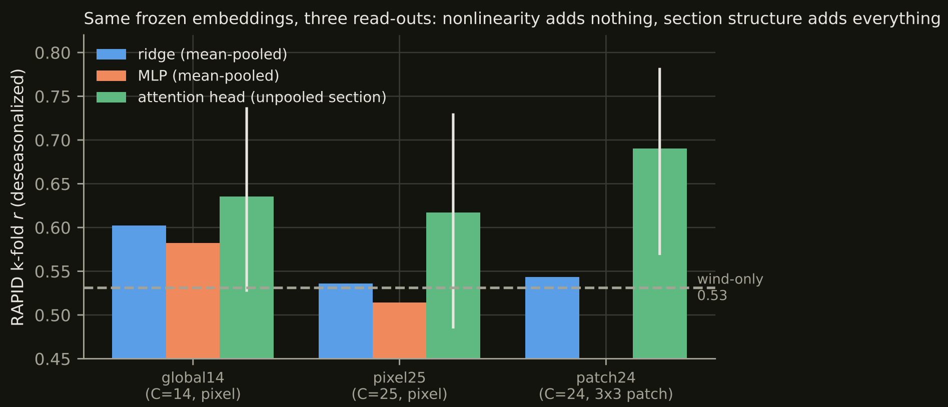

Figure 3: Same frozen embeddings, three read-outs, same year-blocked folds. The MLP (orange) trails the ridge (blue) everywhere — pointwise nonlinearity recovers nothing. The attention head over the unpooled section (green, with 95% CIs) beats every pooled probe, restores the 25-channel codec to parity with the 14-channel one, and reveals the patch codec — invisible to pooled probes — as the strongest representation in the programme at $r = 0.690$.

7.3 The probe ladder: pooling was hiding the signal

Fig. 3 is this revision's central result, and it changes both a number and a story. The number: the patch codec read by the attention head reaches $r = 0.690$ [0.570, 0.781], RMSE 2.02 Sv (and 0.654 on a second codec seed, below) — against 0.60 for the best pooled-linear configuration, and against a wind-only baseline of 0.531 that the pooled deep-tensor probes had barely cleared. The story: the ladder's two rungs dissociate cleanly. Pointwise nonlinearity (MLP rung) recovers nothing on any target for any codec — so the ridge was never the limit as a *function class*. Spatial structure (head rung) recovers a great deal — so the limit was the *pooling*. Mean-pooling the section computes the one statistic in which an east-west difference cancels, and thermal wind says the transport *is* an east-west difference. The two architectural changes of this cycle turn out to be halves of one physical idea: the 3×3 patch puts local gradients into each embedding, the attention head reads gradients across the section, and each looked worthless without the other — the patch codec's pooled numbers were indistinguishable from its pixel control.

The attribution matrix. An end-to-end baseline completes the picture: the *same* attention head, same folds, same regularization, fed raw section anomalies (values + observed-flags) instead of embeddings, at both receptive fields:

tokens carry	raw data	codec embedding
single pixel	0.613 [0.48, 0.73]	0.617 [0.49, 0.73]
3×3 neighbourhood	0.659 [0.55, 0.75]	0.690 / 0.654 (2 seeds)

A second seed, and a retraction. The 0.690 cell was replicated on an independently trained codec of identical configuration. It came back at 0.654 [0.543, 0.748]. The capability claim survives — both seeds beat every pooled probe and the wind-only baseline decisively, so unpooling is real — but the *pretraining margin does not.* The raw-3×3 baseline is codec-independent at 0.659; the two-seed embedding mean is 0.672; the advantage is therefore +0.013, which is nothing. We previously reported +0.031 "positive in direction, not significant" and we withdraw even that directional reading: it was one seed's noise. Note that the pooled ridge replicated far more tightly across the same two codecs (0.543 vs 0.531) than the head did (0.690 vs 0.654) — the more expressive probe is also the noisier instrument, and no future head number in this programme should be quoted from a single seed.

What remains, then, decomposes as: *unpooling* the section is the whole of the improvement (pooled ridge 0.54 → head 0.65+); the 3×3 *receptive field* adds ∼+0.045 and does so on raw data alone; *pretraining* adds nothing measurable at either receptive field. For the monthly RAPID probe, a supervised head over raw section data recovers everything the codec offers. The pretraining case now rests entirely on the scoreboards where 240 labelled months have no answer: field prediction (stage 2 beats persistence everywhere; the matching end-to-end field-forecaster baseline is the open follow-up), transfer to sparsely-labelled targets like MOVE, and whether a margin appears at all once the codec is sized to its Chinchilla anchor.

That last test has now run, on the quarter-degree tensor at 40.7M parameters (§7.4), and it neither rescues the pretraining case nor closes it: the head reads 0.662 [0.557, 0.745] from the codec against 0.628 [0.514, 0.729] from raw 3×3 tokens on the same section, a margin of +0.034 where the 1° pair gave +0.013. Same sign, roughly twice the size, still a small fraction of its own confidence interval, and — the part that governs how it should be quoted — *one seed.* We wrote above that no head number in this programme should be quoted from a single seed, and we decline to exempt this one because it happens to point the way we would prefer.

Honesty box: the head is the most expressive probe we run (∼25k parameters against ∼220 training months per fold), and although it is regularized hard, early-stopped on a train tail, and seed-averaged behind the same blocked folds as every other number, the head-vs-ridge gaps have not yet been given a paired significance test. The second-seed replication is no longer outstanding — it is reported above, and it cost us the pretraining margin. A capacity sweep of the head itself (23k→325k→2.0M parameters) shows *neither* side is parameter-limited: at 325k both columns drop (raw-3×3 0.659 → 0.590, embedding-3×3 0.690 → 0.611), as the label-count arithmetic predicts (∼220 training months per fold support a head of tens of thousands of parameters, not hundreds), and the embedding margin persists at both sizes. The 2.0M cells are running as a formality.

7.4 First light from the quarter-degree tensor

The first codec trained on the quarter-degree tensor — a pilot-size (0.92M) control, C=39, 84,405 pixels, single seed — posts the highest pooled-ridge k-fold of the programme: RAPID $r = 0.620$ [0.484, 0.741], RMSE 2.25 Sv, 18-month low-passed 0.747, 46% of the 2009–10 dip. Three readings, in decreasing confidence. First, the baseline discipline holds: the wind-only ridge measured *on this tensor* is 0.568, so the codec clears its own baseline by +0.052 where the deep-tensor pilots sat exactly on theirs — though the CIs overlap and this is one seed. Second, MOVE transfers again (0.235 [0.112, 0.340], low-passed 0.726) while wind-only at MOVE is *negative* (−0.376): at 16°N the embedding carries signal the wind actively gets wrong. Third, and to be read with care: 0.620 beats every 1° codec on the same truth and protocol, but across tensor generations that sentence means "this data recipe is better", not "this codec is better" (§3.2).

The 41M-parameter codec trained *at* this tensor's compute-optimal anchor has since finished, and its answer is a flat one: pooled k-fold RAPID $r = 0.631$ [0.513, 0.732], RMSE 2.16 Sv, against the pilot's 0.620 on the same tensor and protocol. Forty-four times the parameters bought +0.011, comfortably inside seed noise. Both clear the quarter-degree wind bar (0.568) and neither does so significantly.

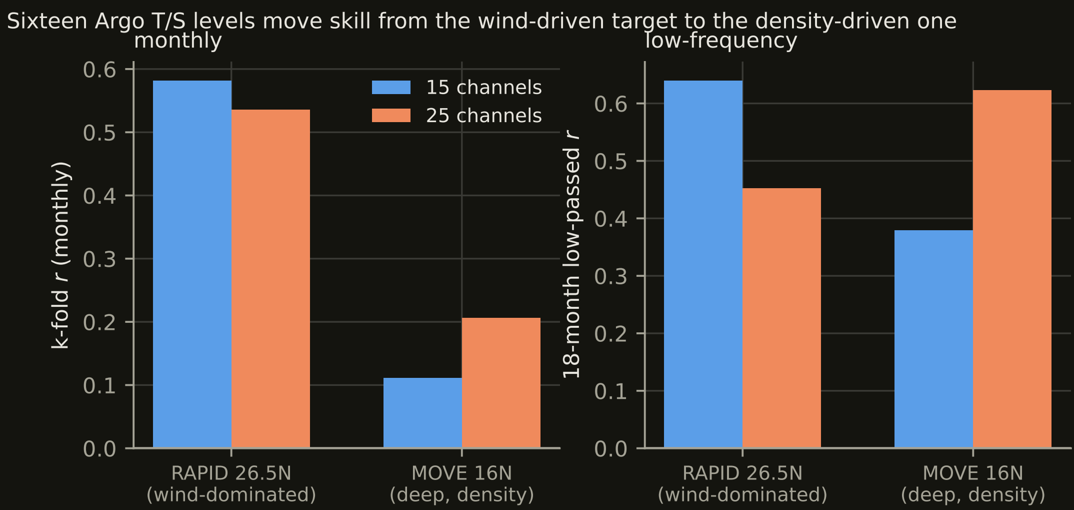

Figure 4: Sixteen Argo temperature and salinity levels to 1900 dbar *appeared* to cost monthly RAPID skill under pooled probes (left, first pair) — resolved by §7.3 as a read-out artifact, not a representation change — while genuinely roughly doubling skill at MOVE (left, second pair) and transforming the low-frequency read-out there (right). MOVE measures the *deep* western-boundary flow at 16°N — precisely the quantity the added levels describe.

The honest reading is that skill on this tensor is *not* capacity-limited at the pooled read-out — which, combined with the flat 50k→1M step sweep of §9, closes both easy scaling axes at once: neither more parameters nor more steps is what stands between this programme and a better transport number. What did move: the unpooled head reads 0.662 [0.557, 0.745] from the same frozen codec, the 2009–10 dip capture rose to **51%** from the pilot's 46%, and the 18-month low-passed correlation reached 0.820 — the last of which should be read against the ∼9 effective degrees of freedom of §5, not as a 0.82-with-240-months would be.

7.5 Where more data moved the skill

Under *pooled* probes, expanding the input appeared to cost RAPID skill: both deep-tensor codecs landed on the wind-only line (0.543 and 0.536 against 0.531) where the coarse tensor held +0.07. Section 7.3 resolves this: the skill had moved into cross-pixel structure, not out of the representation, and the unpooled read-out recovers it in full (0.617–0.690). The RAPID half of this subsection's original claim is therefore retracted; what follows — the MOVE result — is the part that survives, and it was measured pooled, so unpooling can only strengthen it.

At MOVE the same channels do the opposite. The 25-channel codec scores $r = 0.206$ [0.044, 0.346] — the first multi-target result in this programme whose confidence interval excludes zero — with an 18-month low-passed correlation of 0.623, against 0.111 and 0.379 for the 15-channel codec. MOVE is a deep, density-driven transport; RAPID's monthly variance is largely Ekman. The added channels are Argo T and S to 1900 dbar. The skill went where the physics says the information is.

The corrected reading is about read-outs, not bottlenecks: the embedding kept everything, and the probe determined what could be seen. For the generic-embedding hypothesis of §6 this is a *stronger* confirmation than the regional-versus-global comparison — one task-blind representation now serves the wind-dominated and the density-dominated target simultaneously, provided the read-out respects the structure of the question being asked of it.

One more asymmetry is worth recording: the 25-channel codec captures **59%** of the winter 2009–10 collapse out-of-fold, the best of any run in the programme, while its monthly correlation is the second worst. Event anatomy and average correlation are not the same capability, and a table that reports only the latter will hide the former.

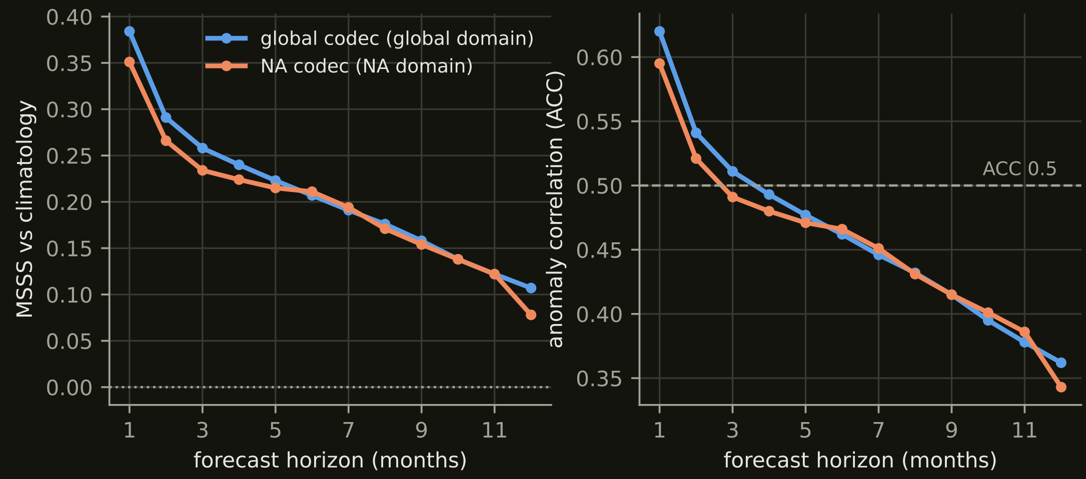

Figure 5: Autoregressive rollout skill versus forecast horizon for the global codec (blue, global domain) and the regional codec (orange, North-Atlantic domain; domains differ, so compare shapes more than levels). Left: MSSS against climatology — still positive at 12 months. Right: anomaly correlation, crossing the conventional 0.5 useful-skill line near four months.

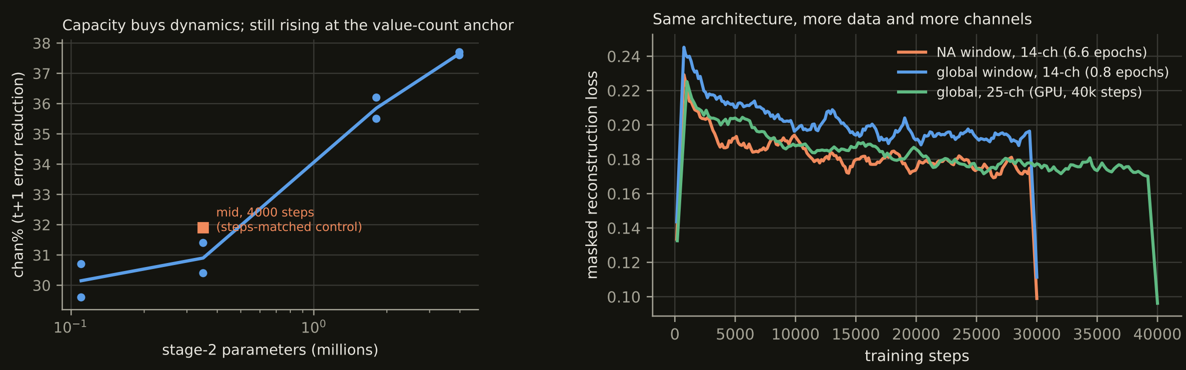

Figure 6: Left: stage-2 capacity sweep (both seeds shown per size) — width×depth buys real dynamics, still rising at 4M parameters; the steps-matched control (orange square) separates capacity from training length. Right: codec training curves across generations.

7.6 How far ahead the ocean can be predicted

Fig. 5: feeding predictions back month after month, the anomaly field stays predictable beyond a year (MSSS +0.11 at h=12; ACC 0.36). The decay is decorrelation, not amplitude damping (the forecast-to-observed amplitude ratio holds near 0.5 throughout). The AMOC read-out on rolled embeddings, by contrast, dies after about a season — transport forecasting at range remains open, consistent with its wind-driven monthly variance.

7.7 Stage-2 capacity and the training curves

7.8 Case study: the winter 2009–10 collapse

8 The static-channel lesson

An ablation found the two climatology input channels contribute *exactly nothing* to any metric: a static map is a function of position, the encoder receives position, and so the channel carries no information the model does not already have. The general rule — a channel is worth what position and season cannot explain of it — now gates every data-acquisition decision. Climatological knowledge is still essential, but it enters as preprocessing arithmetic (the anomaly transform), never as a feature the network must

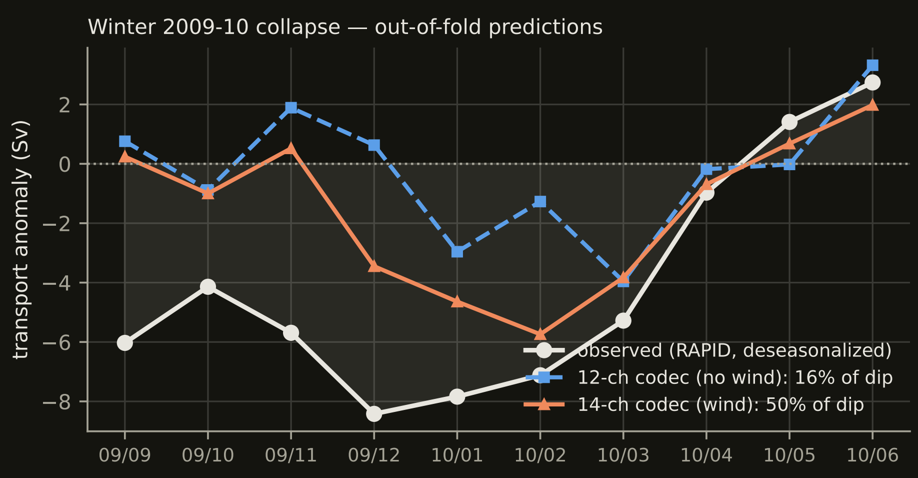

Figure 7: The sharpest event in the RAPID record, predicted out-of-fold. Without wind channels (blue) the model captures 16% of the dip amplitude; with them (orange), 50%. The 25-channel codec reaches 59% (not plotted; see Table 3). The November onset — triggered inside a month — stays invisible to monthly data.

carry through its bottleneck.

9 Limitations and open questions

(1) The wind-only baseline bounds honest claims: monthly AMOC skill is substantially Ekman, though the unpooled head now clears the baseline by +0.10–0.16 where the pooled probes had shrunk the margin to noise. (2) The head result is new and strong, and strength is when scrutiny matters most. The second codec seed has now been run and it removed the pretraining margin (§7.3); what remains outstanding is a paired significance test, and the fact that the head has been run only on RAPID — the MOVE and low-frequency versions may move too. The head is also demonstrably the noisiest probe we own (seed spread 0.036 against the ridge's 0.012), so single-seed head numbers should not be quoted anywhere, including by us. (3) d_z=64 was chosen on a 12-channel tensor and carried unchanged into 24–25 channels; the sweep has not been repeated at the new width. (4) Absolute k-fold values are mildly optimistic — the codec saw non-holdout months' *fields* during self-supervised pretraining (transports were never seen by anything); codec-to-codec comparisons are unaffected. (5) The rollout's horizon is measured only to the holdout-year boundary; the crossover to no skill lies beyond 12 months and is not yet located. (6) Transport forecasting at range is unsolved. (7) *A methodological casualty worth recording*: the stage-2 results for two runs were withdrawn after we found their embedding cache had been populated by a different codec — the shape check (T, P, d_z) matched, so training proceeded happily on the wrong representation, and the signature was healthy embedding-space skill beside catastrophic decoded skill. The cache key now carries a hash of the codec's weights, so a stale cache is a miss rather than a lie. We report the affected numbers as withdrawn rather than regenerating them, because both codecs belong to the superseded coarse tensor.

The training-time ceiling has now been located directly: a 1,000,000-step run at pilot size (LR decayed to a constant 10% floor by 50k steps, probes every 50k) completed in 2.6 h on one RTX 4090 and is *flat from the first probe to the last* — linear r oscillates in $[0.43, 0.48]$, channel-space skill vs. persistence is pinned at 32%, and the final checkpoint's k-fold (0.571) sits within seed noise of the 40k-step run. Twenty-five times the compute of the pilot regime buys nothing at 0.92M parameters: the pilot is capacity-limited, which is what the Chinchilla arithmetic predicted (∼13M parameters for this tensor) and what the 10M-parameter codecs now training are for. Next levers, in expected-value order: paired significance for the head results; the head probe at MOVE and at low frequency; re-running the d_z sweep at C=24; then the approved data ladder — (a) extending the time axis to 1982, where monthly SST and wind exist pre-GLORYS and the Florida cable's 1982–92 decade becomes truth, (b) 0.25° resolution for the GLORYS-derived channels (16× the pixels), (c) doubling the Argo pressure levels — with the Chinchilla arithmetic redone and the codec re-sized at each rung; and label-efficiency curves over the 792 non-RAPID truth months.

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run	window	C	patch	d_z	steps	chan%[†]	k-fold r [95% CI]	dip
coarse tensor (1°, 4 Argo levels)								
pilot4	NA	4	1	32	8k	+29.3	—	—
dz8	NA	12	1	8	30k	+28.6	0.111 [0.01, 0.20]	—
dz16	NA	12	1	16	30k	+30.3	0.151 [0.01, 0.28]	—
dz32	NA	12	1	32	30k	+30.5	0.182 [0.05, 0.31]	—
dz64	NA	12	1	64	30k	+30.6	0.308 [0.13, 0.46]	16%
dz128	NA	12	1	128	30k	+30.2	0.166 [0.07, 0.30]	—
wind14	NA	14	1	64	30k	+35.6	**0.604** [0.47, 0.72]	50%
global14	global	14	1	64	30k	+30.6	**0.602** [0.46, 0.73]	50%
global14b*	global	14	1	64	30k	+30.9	0.556 [0.43, 0.68]	—
global15sst	global	15	1	64	30k	+30.8	0.582 [0.43, 0.71]	47%
deep tensor (1°, 8 Argo levels; §3.2)								
patch24	global	24	**3**	64	40k	+32.0[‡]	0.543 [0.43, 0.66]	27%
pixel25	global	25	1	64	40k	+30.9[‡]	0.536 [0.38, 0.68]	**59%**
pixel24	global	24	1	64	40k	—	0.515 [0.41, 0.63]	—
pixel24-30k	global	24	1	64	30k	—	0.585 [0.45, 0.71]	—
pixel24-60k	global	24	1	64	60k	—	0.549 [0.42, 0.67]	—
patch24-30k	global	24	3	64	30k	—	0.551 [0.42, 0.68]	—
patch24-1M	global	24	3	64	**1M**	+32.1[‡]	0.571 [0.42, 0.70]	—
patch24-seed2	global	24	3	64	40k	—	0.531 [0.40, 0.65]	27%
pixel24-seed2	global	24	1	64	40k	—	0.564 [0.44, 0.69]	—
big10M (10.26M)	global	24	3	64	60k	—	**0.578** [0.45, 0.70]	26%
quarter-degree tensor (0.25° NA, 16 Argo levels, SSH; §7.4)								
f3-pilot	NA 0.25°	39	3	64	40k	—	**0.620** [0.48, 0.74][§]	46%
f3-anchor41M (40.7M)	NA 0.25°	39	3	64	60k	+31.5	0.631 [0.51, 0.73][§]	**51%**

Table 3: **Master table: every codec run.** Generated from each run's own JSON by `ml/make_table.py`, not transcribed. [†]chan% is comparable only within one channel set (its persistence baseline moves with the channels); cross-run comparison goes through the year-blocked k-fold RAPID correlation. Dip = share of the winter 2009–10 collapse amplitude captured out-of-fold. The wind-stress-only ridge baseline (no embedding) scores $r = 0.531$ — the line every codec must beat *on the* 1° *tensors*; the same baseline re-measured on the quarter-degree tensor is 0.568, which is what the [§] row must be read against — never against 0.531. k-fold r here is the POOLED RIDGE rung of the probe ladder (the comparable-across-time instrument); the unpooled attention-head values (§7.3) are higher for every codec measured: global14 0.635, pixel25 0.617, patch24 **0.690** and 0.654 on its second seed, and f3-anchor41M **0.662** [0.56, 0.75] against 0.628 [0.51, 0.73] for the matched raw-3×3 head on the same section — a pretraining margin of +0.034 from one seed, reported here only because the pooled row above it would otherwise be read as the codec's ceiling. *global14b is an exact-configuration replication on a different runner: it measures codec-training seed noise on the k-fold at $\approx \pm 0.05$ — the four pixel24/patch24 rows, spanning 0.515–0.585 across 30k–60k steps, are consistent with that noise and show no step-count trend; the patch24-1M row settles the saturation question directly — every probe metric is flat from the first 50k-step probe to the millionth step (§9), so the pilot size is capacity-limited, not compute-limited. [‡]in-training probe at the final step, not the post-hoc trainprobe used for coarse-tensor rows. Stage-2 figures for global14b and global15sst are *withdrawn*, not stale: both were trained on a poisoned embedding cache (§9).